MILITARY SPECIFICATION

RADIO SET AN/URC-94(V)

This amendment forms a part of Military Specification MIL-R-28848(EC), dated 30 November 1978, and is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.1 "STANDARDS", "MILITARY", after "MIL-STD-109", add:

"MIL-STD-129 Marking For Shipment And Storage".

* 2.1 "STANDARDS", "MILITARY", "MIL-STD-461", title: Delete and substitute: Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference".

* 2.1 "STANDARDS", "MILITARY", after "MIL-STD-810", add:

DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
DoD-STD-1686	Electrostatic Discharge Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)".

* 2.1 "PUBLICATIONS", "National Security Agency", delete "CSESD-32 TSEC KY-58".

PAGE 3

* 3.4, line 2: Delete ", CSESD-32,".

3.4.1.1: Delete.

3.4.2.2, line 2: Delete "0.5" and substitute "1.0".

3.4.2.2, line 3: Delete "1.0" and substitute "3.0".

PAGE 4

* After "3.4.3.1", add the following new paragraphs:

"3.4.3.2 Electrostatic discharge (ESD). When the contract requires design or redesign, electrostatic protection shall be provided in accordance with the Design protection paragraph of DoD-STD-1686.

3.4.3.3 ESD marking.

3.4.3.4 Assemblies. Assemblies shall be marked in accordance with DoD-STD-1686.

3.4.3.5 Equipment enclosures. Equipment enclosures shall be marked in accordance with DoD-STD-1686.

3.4.3.6 Nonstandard parts. Electrostatic discharge sensitive (ESDS) nonstandard parts shall be marked with the sensitive electronic device symbol specified in MIL-STD-129.".

AMSC N/A FSC 5820

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:50Z
Check the source to verify that this is the current version before use.

* 3.4.4: Delete and substitute:

"3.4.4 Safety. The requirements of the Safety criteria paragraph of MIL-E-16400 shall be incorporated in the design of the equipment to preclude personnel from accidentally coming into contact with hazardous electrical potentials.".

* 3.4.5: Delete and substitute:

"3.4.5 Electromagnetic compatibility (EMC). The equipment shall be self-compatible and shall conform to the requirements of MIL-STD-461 for Class A3 and A4 equipments. MIL-HDBK-241 may be used for guidance in electromagnetic interference (EMI) design of power supplies.".

3.5.1.1, line 2: Delete "20.32 cm (8.0 in)" and substitute "22.86 cm (9.0 in)".

PAGE 5

* 3.5.3.1.1: Delete and substitute:

"3.5.3.1.1 AC power source. 115/230 volts AC ±10 percent single-phase 47 to 63 hertz (Hz), including 60-Hz, Type I, shipboard power, as specified in DoD-STD-1399, Section 300. The equipment shall comply with the user equipment requirements and constraints of DoD-STD-1399 when using Type I power.".

* 3.5.3.7.1: Delete title and substitute: "HF coupler primary power".

PAGE 6

3.5.4.2, line 1: Between "enclosure" and "shall", insert "for the RT-1230(V)-1/URC-94(V)".

PAGE 8

TABLE II: In the Indicator column, after "Power meter", add "(with a red lens)".

PAGE 11

3.5.6.3.3.1, line 2: Delete "350" and substitute "300".

PAGE 12

3.6.2.3, lines 1 and 2: Delete "exposed equipment." and substitute "equipment utilized in a sheltered area.".

PAGE 15

3.6.5.10.2: Delete the text and substitute "During normal operation, the coupler shall provide Minimum Detrimental Effect on a receiver signal from the antenna to the 50-ohm coaxial cable from the associated receiver/transmitter, due to the untuned state of the matching network, over a signal that would be present if no coupler were used.".

3.8.1, lines 4 and 5: Delete "of the associated RT-1230(V)/URC-94(V)" and substitute "for the amplifier/control unit. A muting circuit to disable the speaker of a second amplifier/control unit shall be provided.".

PAGE 16

3.9.1, lines 1 and 2: Delete "20.32 cm (8.0 in)" and "11.43 cm (4.5 in)" and substitute "22.52 cm (9.25 in)" and "13.33 cm (5.25 in)", respectively.

* 3.12, (added by AMENDMENT 1): Delete and substitute:

"3.12 Workmanship screen. All equipments produced in accordance with this specification shall withstand a defect detection vibration screen of random vibration at $0.02g^2$/Hz ±3 dB from 80 Hz to 350 Hz and the temperature cycling specified in 4.9.2."

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:50Z
Check the source to verify that this is the current version before use.

PAGE 17

TABLE IV, in the "Inspection or test" and "Requirement paragraph" columns: Delete "Equipment conditioning" and "3.4.1.1" and substitute "Workmanship screen" and "3.12", respectively.

TABLE IV, on the line with "Workmanship": Add "X" in the "FA" and "p" columns.

TABLE IV, on the lines with "Temperature ranges" and "Vibration": Delete "X" in the "E" column.

PAGE 22

* 4.9.2, 4.9.2.1, 4.9.2.2, 4.9.2.3, and 4.9.2.4 (added by AMENDMENT 1): Delete and substitute:

"4.9.2 Workmanship screen. Vibration and temperature cycling shall be performed on each equipment. Vibration shall be performed prior to temperature cycling. The vibration may be performed at the module, drawer, or end item level. All the hardware including cables and connectors shall be exposed to vibration. The workmanship screen shall be the first inspection performed under production inspection.

"4.9.2.1 Random vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 2. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

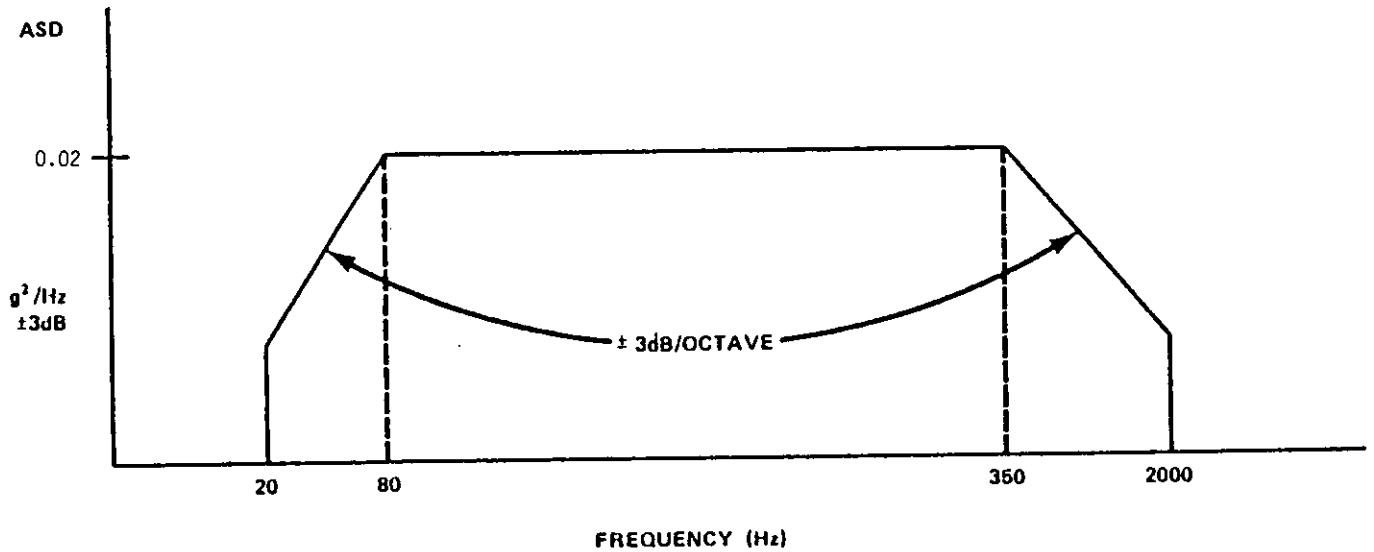


FIGURE 2. Random vibration curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:50Z
Check the source to verify that this is the current version before use.

"4.9.2.2 Temperature cycling. Each unit shall be subjected to 10 cycles of the temperature curve shown in FIGURE 3. The temperature rate of change shall be not less than 5°C per minute. Unit power shall be turned on and off at the indicated times. The unit shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the unit reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles.

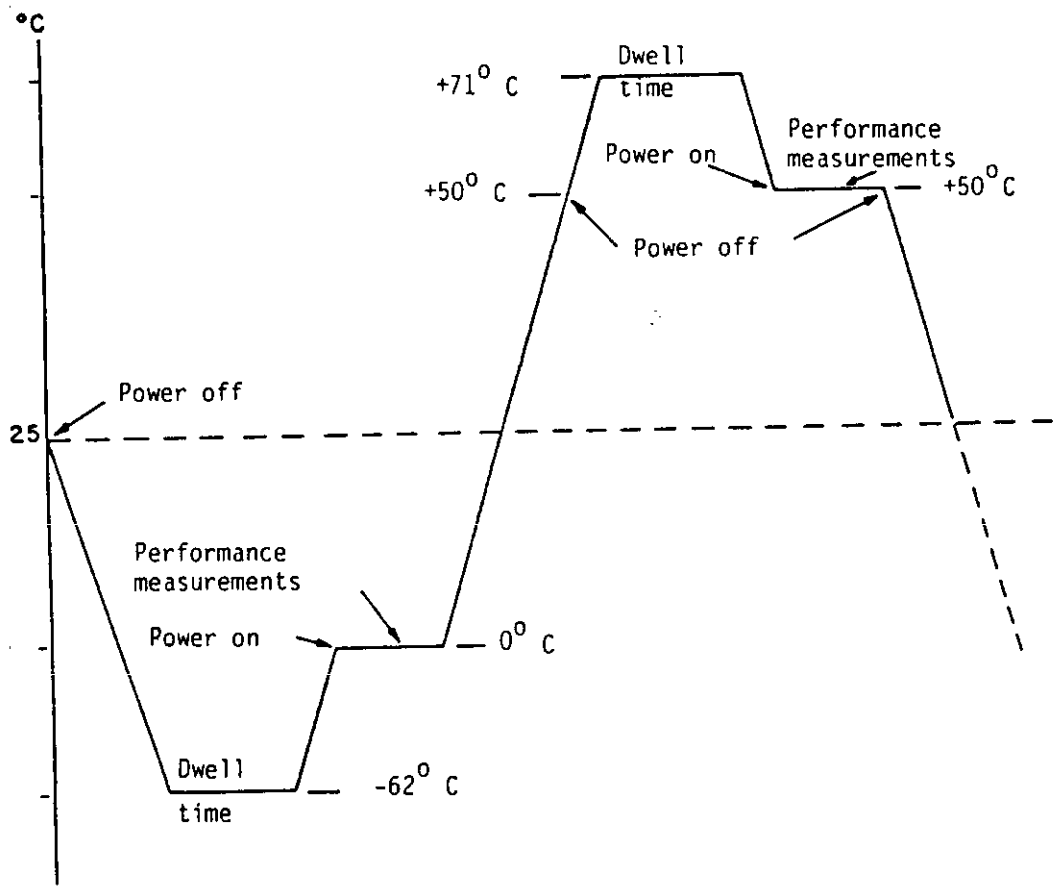


FIGURE 3. One cycle of temperature curve (not to scale)".

* Add the following new paragraph:

"4.9.2.3 Equipment configuration. Individual equipments (as specified in 3.3) need not be tested in a system configuration provided provision is made to fully exercise all active electrical and electromechanical elements. The antenna and H-189/U handset may be excluded from the test.".

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:50Z
Check the source to verify that this is the current version before use.

PAGE 24

"4.9.3, after the last sentence: Add "The resulting data shall serve as baseline for tests to be conducted as specified in 4.4.3.".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
NAVY - EC

(Project No. 5820-N815(EC))

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:50Z
Check the source to verify that this is the current version before use.